

Knapsack Program

Saturday, 7 February 2026

7:34 PM

P

0	1	2	3	4
0	1	2	5	6

 $n=4$

w

0	1	2	3	4
0	2	3	4	5

 $m=8$

K	w	0	1	2	3	4	5	6	7	8
0	0	0	0	0	0	0	0	0	0	0
1	0	0	1	1	1	1	1	1	1	1
2	0	0	1	2	2	3	3	3	3	3
3	0	0	1	2	5	5	6	7	7	7
4	0	0	1	2	5	6	6	7	8	8

Now to check which one to choose once we have the data in above table.

$i=n$; $j=m$
 $i=4$; $j=8$
 while ($i>0$ & $j>0$)
 if ($k[i][j] == k[i-1][j]$)

if $\hookrightarrow$ value present in above row too.

cout << i << "\n" << endl

$i--$;

}

else

if

... L ... 0 ... 4 ...

main()

{

int $P[5] = \{0, 1, 2, 5, 6\}$

int $wt[5] = \{0, 2, 3, 4, 5\}$

int $m=8, n=4$;

int $k[5][9]$;

for (int $i=0$; $i \leq n$; $i++$)

{

for (int $w=0$; $w \leq m$; $w++$)

{

① if (int $i==0$ || $w==0$)

$k[i][w] = 0$;

else if ($wt[i] \leq w$)

② $k[i][w] = \max (P[i] + k[i-1][w-wt[i]], k[i-1][w]);$

else

③ $k[i][w] = k[i-1][w];$

}

cout << $k[n][m]$;

```
while (i <= 1) <endl;
{ i--;
  j = j - wt[i];
}
```